University of Copenhagen
Faculty of Science
Department of Chemistry

Optimized Virtual Orbital Space for Quantum Computing

Master's Thesis in Chemistry / Quantum Chemistry

Author
Tobias ⟨*Surname*⟩
xxx@ku.dk

Supervisor
⟨Supervisor Name⟩
Department of Chemistry, UCPH

Co-supervisor (if any)
⟨Co-supervisor Name⟩

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Abstract

⟨*Write the abstract here after all chapters are complete.*⟩

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Resumé

⟨*Danish translation of the abstract.*⟩

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List of Abbreviations

HF	Hartree-Fock
RHF	Restricted Hartree-Fock
UHF	Unrestricted Hartree-Fock
MP2	Second-Order Møller-Plesset Perturbation Theory
MPn	Møller-Plesset Perturbation Theory of order n
CI	Configuration Interaction
CISD	Configuration Interaction Singles and Doubles
FCI	Full Configuration Interaction
CC	Coupled Cluster
CCSD	Coupled Cluster Singles and Doubles
MCSCF	Multi-Configurational Self-Consistent Field
CASSCF	Complete Active Space Self-Consistent Field
DFT	Density Functional Theory
OVOS	Optimized Virtual Orbital Space
FNO	Frozen Natural Orbitals
PNO	Pair Natural Orbitals
DLPNO	Domain-Based Local Pair Natural Orbitals
NO	Natural Orbitals
VQE	Variational Quantum Eigensolver
QPE	Quantum Phase Estimation
JW	Jordan-Wigner
BK	Bravyi-Kitaev
NISQ	Noisy Intermediate-Scale Quantum
QC	Quantum Computing
BO	Born-Oppenheimer
AO	Atomic Orbital
MO	Molecular Orbital

SCF	Self-Consistent Field
LCAO	Linear Combination of Atomic Orbitals
ERI	Electron Repulsion Integral
CBS	Complete Basis Set
GTO	Gaussian-Type Orbital
STO	Slater-Type Orbital
DIIS	Direct Inversion in the Iterative Subspace
NR	Newton-Raphson
RLE	Reduced Linear Equation
PySCF	Python-based Simulations of Chemistry Framework
UCPH	University of Copenhagen

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Chapter 1

Introduction

1.1 Motivation

1.2 The Virtual Orbital Space Problem

1.3 Quantum Computing Context

1.4 Objectives

1.5 Thesis Outline

Chapter 2

Background

This chapter provides the necessary theoretical background for understanding the electronic structure problem, Hartree-Fock theory, molecular orbital integrals, Møller-Plesset perturbation theory, and the spin-orbital formalism. It also introduces the orbital space partitioning and index notation conventions that are used consistently throughout this thesis. Finally, it briefly discusses the relevance of orbital space in the context of quantum computing algorithms for chemistry. Each section builds on the previous ones to establish a solid foundation for the development of the OVOS method in Chapter 3.

2.1 The Electronic Structure Problem

The electronic structure problem is the task of solving the non-relativistic, time-independent Schrödinger equation for a system of N electrons moving in the electrostatic potential of M fixed nuclei, reflecting the Born-Oppenheimer approximation. Typically, the electronic Hamiltonian is expressed as:

$$\hat{H}_{el}\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$$

where $\hat{H}_{el}$ is the electronic Hamiltonian, Ψ is the many-electron wavefunction, and E is the electronic energy. The electronic Hamiltonian can be written as:

$$\hat{H}_{el} = \sum_{i=1}^{N}\left(-\frac{1}{2}\nabla_i^2 - \sum_{A=1}^{M}\frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|}\right) + \sum_{i<j}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

where the first term represents the kinetic energy of the electrons, the second term is the nuclear attraction, and the third term is the electron-electron repulsion.

The electron-electron repulsion term couples all electrons together, making the exact solution of the Schrödinger equation computationally intractable for all but the smallest systems. This motivates the development of approximate methods that can capture the essential physics of electron correlation while remaining computationally feasible.

Hartree-Fock theory provides a mean-field approximation to the electronic structure problem. The residual correlation energy, defined as $E_{corr} = E_{exact} - E_{HF}$, quantifies the error of the Hartree-Fock approximation, accurately capturing only a fraction of the total correlation energy.

For small molecules, the correlation energy can on the order of tenths of a Hartree, and can be critical for accurately describing chemical phenomena such as bond dissociation and the properties of open-shell molecules. The following section introduces Hartree-Fock theory in more detail, setting the stage for the post-Hartree-Fock methods that are the focus of this thesis.

Placeholder

Figure 2.1: Potential energy curves for H_2 illustrating the failure of HF near dissociation and the importance of electron correlation.

2.2 Hartree-Fock Theory

Hartree-Fock (HF) theory approximates the N-electron wavefunction as a single Slater determinant of N orthonormal spin-orbitals, treating electron-electron repulsion in a mean-field sense. Despite its simplicity, HF underpins virtually all post-HF correlated methods.

Slater determinants are described by the Pauli exclusion principle, ensuring that no two electrons can occupy the same quantum state. This antisymmetry is crucial for accurately describing fermionic systems. The variational principle leads to the Fock equations, which can be expressed in matrix form as the Roothaan-Hall equations: $FC = SC\epsilon$, where F is the Fock matrix, S is the overlap matrix, C is the coefficient matrix for the molecular orbitals, and ϵ is the diagonal matrix of orbital energies. The Fock matrix incorporates the Coulomb (J) and exchange (K) operators, which represent the average electron-electron repulsion and the antisymmetry of the wavefunction, respectively. It is this equation that is solved iteratively in the self-consistent field (SCF) procedure, often accelerated by techniques that expedite convergence, like DIIS. The variational principle ensures that the HF energy is an upper bound to the exact ground state energy, $E_{HF} = \langle \Phi_{HF} | \hat{H} | \Phi_{HF} \rangle \geq E_{exact}$. Which allows us to define that the correlation energy is negative: $E_{corr} = E_{exact} - E_{HF} < 0$. The resulting orbital energies ϵ_p provide insight into the electronic structure, with occupied orbitals typically having negative energies and virtual orbitals having positive energies, as per Koopmans' theorem.

2.2.1 Restricted and Unrestricted Formalism

HF varying in its treatment of spin leads to two different formalisms: Restricted Hartree-Fock (RHF) and Unrestricted Hartree-Fock (UHF). Spin orbitals are constructed from

spatial orbitals and spin functions,

$$\chi_p(\mathbf{r}, \sigma) = \phi_p(\mathbf{r})\alpha(\sigma) \quad \text{or} \quad \chi_p(\mathbf{r}, \sigma) = \phi_p(\mathbf{r})\beta(\sigma)$$

where α and β are the spin functions for spin-up and spin-down electrons, respectively. RHF assumes that each spatial orbital, ϕ_p, is doubly occupied by electrons of opposite spin, making it suitable for closed-shell systems. Translating to spin-orbital language, this means that the same spatial orbital is used for both α and β electrons, leading to a single set of molecular orbitals. RHF keeps the total spin of the system fixed, ensuring that the wavefunction is an eigenfunction of the total spin operator $\hat{S}^2$. However, RHF fails to describe open-shell systems and bond dissociation correctly, as it cannot capture the correct spin symmetry in these cases. In contrast, UHF allows for different spatial orbitals for α and β spins, making it applicable to open-shell systems. UHF allows for spin deviation from pure spin states, leading to spin contamination where the expectation value of the total spin operator $\langle \hat{S}^2 \rangle$ deviates from the expected value for a pure spin state. In bond breaking scenarios, UHF can provide a qualitatively correct description of the dissociation process, while RHF fails to capture the correct physics. For this reason, UHF is used throughout this thesis for generality, as it can handle both closed-shell and open-shell systems effectively. It is recognized that it comes with an increased computational cost compared to RHF. RHF and UHF have similar scaling with respect to the number of basis functions, but UHF takes on the additional degrees of freedom in the spin orbitals.

2.3 Molecular Orbital Integrals

At the core of electronic structure methods are the one- and two-electron integrals evaluated in the molecular orbital basis. These integrals form the building blocks of the Roothaan-Hall equations and the correlation energy expressions used in post-HF methods. This section defines these integrals and establishes the notation conventions applied consistently throughout this thesis, resolving an important ambiguity that arises when comparing PySCF output with the formulas of Adamowicz & Bartlett.

2.3.1 One-Electron Integrals

One-electron integrals reflect the kinetic energy of electrons and their attraction to the nuclei. In an atomic orbital (AO) basis, the overlap integral is defined as: $S_{\mu\nu} = \langle \mu | \nu \rangle = \int \phi_\mu^*(\mathbf{r})\phi_\nu(\mathbf{r})d\mathbf{r}$, the kinetic energy integral is: $T_{\mu\nu} = \langle \mu | -\frac{1}{2}\nabla^2 | \nu \rangle$, and the nuclear attraction integral is: $V_{\mu\nu} = \langle \mu | -\sum_A \frac{Z_A}{|\mathbf{r}-\mathbf{R}_A|} | \nu \rangle$. The core Hamiltonian in the AO basis is then given by: $h_{\mu\nu} = T_{\mu\nu} + V_{\mu\nu}$. To transform these integrals to the molecular orbital (MO) basis, we use the MO coefficients from the HF calculation: $h_{pq} = \sum_{\mu\nu} C_{\mu p}^* h_{\mu\nu} C_{\nu q}$, where C is the coefficient matrix that transforms from the AO to MO basis. This transformation is essential for evaluating the Fock matrix and the correlation energy in the MO basis. The one-electron integrals are straightforward to compute and are typically stored in a dense matrix format.

2.3.2 Two-Electron Integrals

Two-electron integrals, also known as electron repulsion integrals (ERIs) are defined as: $(pq|rs) = \int\int \phi_p^*(1)\,\phi_q(1)\,\frac{1}{r_{12}}\,\phi_r^*(2)\,\phi_s(2)\,d1\,d2$ in chemists' notation, which is the convention used by PySCF. These integrals represent the Coulombic repulsion between pairs of electrons and are central to the construction of the Fock matrix and the evaluation of correlation energies. The ERIs exhibit an 8-fold permutation symmetry for real orbitals, which can be exploited to reduce computational cost. This allows for the storage of only unique integrals, for indexes $p \geq q$ and $r \geq s$, and $(pq|rs) = (qp|rs) = (pq|sr) = (qp|sr) = (rs|pq) = (sr|pq) = (rs|qp) = (sr|qp)$. The Fock matrix in the MO basis can be expressed in terms of these integrals as: $F_{pq} = h_{pq} + \sum_i [2(pi|qi) - (pq|ii)]$ for RHF, where the sum runs over occupied orbitals. The two-electron integrals are typically stored in a 4-dimensional array, and their evaluation is the computational bottleneck in many electronic structure methods. For UHF calculations, the Fock matrix has separate contributions from α and β electrons, leading to a larger number of integrals to be evaluated and stored.

2.3.3 Integral Notation: Chemists' vs. Physicists' Conventions

The notation for two-electron integrals can differ between chemists and physicists, leading to potential confusion when comparing formulas from the literature with the output of quantum chemistry software. In chemists' notation, the integral is written as $(pq|rs)$, while in physicists' notation, it is written as $\langle pq|rs\rangle = (pr|qs)$. The antisymmetrised integral, which is relevant for post-HF methods, is defined as $\langle pq\|rs\rangle = \langle pq|rs\rangle - \langle pq|sr\rangle = (pr|qs) - (ps|qr)$. The OVOS formulas in this thesis use the antisymmetrised physicists' notation, while PySCF returns integrals in the chemists' notation. The conversion between these notations is explicitly implemented in the code as: $\langle pq\|rs\rangle = (pr|qs) - (ps|qr)$. This establishes a clear and consistent framework for interpreting the integrals used in the OVOS method and ensures that the theoretical formulas align with the computational implementation.

2.4 Møller-Plesset Perturbation Theory

Møller-Plesset perturbation theory (MPPT) treats electron correlation as a perturbation on top of the mean-field Fock operator. Truncated at second order (MP2), it is the simplest post-HF method that captures dynamical correlation at $\mathcal{O}(N^5)$ computational cost, and furnishes the energy functional that OVOS seeks to optimise.

2.4.1 MP2 Correlation Energy

The MP2 correlation energy is derived by partitioning the electronic Hamiltonian as $\hat{H} = \hat{H}_0 + \lambda\hat{V}$, where $\hat{H}_0$ is the Fock operator, which is diagonal in the canonical molecular orbital basis, and $\hat{V}$ is the fluctuation potential that accounts for electron correlation.

The zeroth-order wavefunction is the Hartree-Fock determinant $|\Phi_0\rangle$, and Brillouin's theorem ensures that there are no first-order contributions from singly-excited determinants.

The first-order wavefunction is therefore a linear combination of doubly-excited determinants, and the MP1 amplitudes are given by: $t_{ij}^{ab} = -\frac{\langle ab\|ij\rangle}{\varepsilon_a+\varepsilon_b-\varepsilon_i-\varepsilon_j}$, where $\langle ab\|ij\rangle$ is the antisymmetrised two-electron integral in physicists' notation.

The MP2 correlation energy is then computed as: $E_{corr}^{(2)} = \frac{1}{4}\sum_{i,j,a,b} t_{ij}^{ab}\langle ij\|ab\rangle$, where the sum runs over all occupied orbitals i, j and virtual orbitals a, b. In practice, the summation can be restricted to $i > j$ and $a > b$ to avoid double counting, leading to: $E_{corr}^{(2)} = \sum_{i>j,a>b} t_{ij}^{ab}\langle ij\|ab\rangle$. It is critical to make the distinction that this expression sums over the entire virtual orbital space, which is the full MP2 energy. In section 2.6, we will contrast this with the truncated MP2 energy used in OVOS, where the summation is restricted to a subset of active virtual orbitals.

2.4.2 High-Order Amplitudes and T1 definition

In MP2, the MP1 amplitudes correspond to double electron excitations (T2 amplitudes). The T1 amplitudes, which represent single excitations, do not appear in the MP2 energy expression due to Brillouin's theorem, which states that the matrix elements of the Hamiltonian between the Hartree-Fock determinant and singly-excited determinants vanish. The presence of non-zero T1 amplitudes is an indicator that the reference determinant is not optimal, as it suggests that the orbitals can be rotated to lower the energy. In the context of coupled cluster theory, the stationarity condition $\nabla_R E(R) = 0$ leads to the definition of Brueckner orbitals, which are those for which the T1 amplitudes vanish. This connection between orbital optimization and the nature of the excitation amplitudes is a key motivation for developing methods like OVOS that seek to optimize the virtual orbital space to minimize correlation energy.

2.4.3 Spin-Orbital Formulation

In the unrestricted Hartree-Fock (UHF) formalism, the indices i, j run over all α and β occupied orbitals, while a, b run over all α and β virtual orbitals. The antisymmetrised integral form unifies the treatment of $\alpha\alpha$, $\beta\beta$, and $\alpha\beta$ pairs, and the spin-orbital Fock matrix is block-diagonal, with separate blocks for α and β spins. The off-diagonal Fock elements vanish for canonical orbitals, ensuring that the orbital energies ε_p are diagonal in the spin-orbital basis.

2.4.4 MP2 Limitations and Applicability

MP2 is size-consistent, meaning that the correlation energy of non-interacting subsystems is additive, $E_{corr}^{(2)} = \sum_{i>j,a>b} t_{ij}^{ab}\langle ij\|ab\rangle$, which is a crucial property for accurately describing large systems and reaction energies. For a truncated MP2 calculation, where the summation is restricted to a subset of orbitals. However, MP2 is a single-reference method and can fail in cases where the reference determinant is not a good approximation to the true wavefunction, such as in bond dissociation or transition states. In such cases, higher-order methods or multi-reference approaches may be necessary to capture the correct physics.

2.4.5 Computational Scaling Motivation

The $\mathcal{O}(N^5)$ computational scaling of MP2 presents a significant barrier for large systems and higher-order methods. The Optimized Virtual Orbital Space (OVOS) method, detailed in Chapter 3, applies a strategic solution: by restricting the correlation energy calculation to a small, actively-correlated subset of virtual orbitals, the effective scaling can be reduced to $\mathcal{O}(N_{occ}^2 {N'_{virt}}^2)$ or better, where $N'_{virt} \ll N_{virt}$. This active space reduction is particularly powerful when combined with higher-order methods like CCSD(T), which scale as $\mathcal{O}(N^6)$ or worse in the full space.

2.5 Orbital Space

A quantum state in electronic structure theory is described by a wavefunction that depends on the coordinates of all electrons. Each quantum state, which can be represented by Slater determinants, is constructed from a set of spin-orbitals, which are themselves linear combinations of atomic orbitals:

$$\chi_p(\mathbf{r}, \sigma) = \phi_p(\mathbf{r})\alpha(\sigma) = \sum_\mu C_{\mu p}\phi_\mu(\mathbf{r})\sigma(\sigma)$$

where ϕ_μ are the atomic orbitals, $C_{\mu p}$ are the MO coefficients, and σ is the spin function.

The spatial part of the spin-orbitals come from basis sets, which are typically composed of Gaussian-type orbitals (GTOs) or Slater-type orbitals (STOs). The choice of basis set can significantly affect the accuracy and computational cost of electronic structure calculations. Commonly used basis sets include minimal basis sets like STO-3G, split-valence basis sets like 6-31G, and correlation-consistent basis sets like cc-pVDZ. The size and quality of the basis set determine the number of spin-orbitals available for constructing the wavefunction, which in turn affects the description of electron correlation and the convergence of post-HF methods. In this thesis, we use 6-31G and cc-pVDZ basis sets for our test calculations, as they provide a good balance between accuracy and computational cost of the system.

2.5.1 Partitioning and Index Notation

Orbitals in electronic structure theory are often partitioned into different subspaces to facilitate the formulation of post-HF methods. The Optimised Virtual Orbital Space (OVOS) method partitions the full spin-orbital basis into three non-overlapping subsets: occupied, active unoccupied (active virtual), and inactive unoccupied (inactive virtual). The occupied orbitals, denoted by indices i, j, k, l, correspond to the spin-orbitals that are occupied in the Hartree-Fock reference sate, and they are indexed from 0 to $n_{elec} - 1$, where n_{elec} is the number of electrons. The active virtual orbitals, denoted by indices a, b, c, d, are the subset of unoccupied orbitals that are optimised in the OVOS method, and they are indexed from n_{elec} to $n_{elec} + N'_{virt} - 1$, where N'_{virt} is the number of active virtuals and is a key parameter in the OVOS method. The inactive virtual orbitals, denoted by indices e, f, g, h, are the remaining unoccupied orbitals that are not explicitly rotated in the OVOS method; they define the complement space and are indexed from $n_{elec} + N'_{virt}$ to $n_{spin} - 1$, where n_{spin} is the total number of spin-orbitals. The OVOS rotation operates only in the block that couples the active virtuals with the inactive

virtuals, while the occupied orbitals remain fixed. The full set of virtual orbitals can be denoted as $a..e$, which includes all orbitals from n_{elec} to $n_{spin} - 1$ for a full MP2 calculation. This partitioning is illustrated in Figure 2.2, and the index notation is summarised in Table 2.1.

Placeholder

Figure 2.2: Partitioning of spin-orbital space for OVOS. The dashed line separates the active and inactive virtual subspaces being optimised.

Label	Indices	Orbital Space	Range	Count
Occupied	i, j, k, l	occ.	$[0, n_{elec})$	n_{elec}
Active Virtual	a, b, c, d	act. virt.	$[n_{elec}, n_{elec} + N'_{virt})$	N'_{virt}
Inactive Virtual	e, f, g, h	inact. virt.	$[n_{elec} + N'_{virt}, n_{spin})$	$N_{virt} - N'_{virt}$
All Virtual	$a..h$	full virt.	$[n_{elec}, n_{spin})$	N_{virt}

Table 2.1: Orbital subspace notation used throughout this thesis.

By this partitioning scheme, the OVOS method focuses on optimising the active virtual subspace to minimize the correlation energy, while the inactive virtuals provide a fixed reference that defines the complement space. The truncated MP2 energy formula used in OVOS is given by: $E_{corr,trunc} = \sum_{i>j\in occ, a>b\in act.virt.} t_{ij}^{ab} \langle ij \| ab \rangle$, where the summation is restricted to the active virtuals, and the inactive virtuals do not appear in this explicit correlation energy sum. This truncation reduces the computational cost from $\mathcal{O}(N^5)$ to $\mathcal{O}(N'^5)$, where N' is the number of active virtuals, which is typically much smaller than the total number of virtuals. The choice of N'_{virt} is a critical aspect of the OVOS method, as it involves a trade-off between accuracy and computational cost.

2.6 Spin-Orbital Formalism

The spin-orbital formalism provides a framework for unrestricted calculations by treating each spatial orbital as giving rise to a pair of spin-orbitals, one for each spin projection

(α and β). A spin-orbital is defined as $\chi_p(x) = \phi_p(\mathbf{r})\sigma(s)$, where $\phi_p(\mathbf{r})$ is the spatial part of the orbital and $\sigma(s)$ is the spin function, which can be either α (spin-up) or β (spin-down). The total number of spin-orbitals is therefore $n_{spin} = 2 \times n_{spatial}$. In this implementation, an interleaved ordering convention is used for the spin-orbitals: even indices correspond to α spin-orbitals, while odd indices correspond to β spin-orbitals, as illustrated in Figure 2.3. The total number of spatial orbitals is $n_{spatial} = n_{spin}/2$, and the mapping is as follows: index 0 corresponds to spatial orbital 0 with α spin, index 1 corresponds to spatial orbital 0 with β spin, index 2 corresponds to spatial orbital 1 with α spin, etc.

This interleaved ordering allows for a unified treatment of α and β electrons in the construction of the Fock matrix and the evaluation of integrals. The Fock matrix in the spin-orbital basis is block-diagonal, with separate blocks for α and β spins. $F_{spin} = \begin{pmatrix} F_\alpha & 0 \\ 0 & F_\beta \end{pmatrix}$ where F_α and F_β are the Fock matrices for the α and β spin-orbitals, respectively.

The two-electron integrals in the spin-orbital basis are also constructed according to this interleaved ordering, with specific blocks corresponding to different spin combinations (e.g., $\alpha\alpha$, $\beta\beta$, $\alpha\beta$). After the construction of the Fock matrix and the integrals in the spin-orbital basis, the spin-orbitals are re-sorted by their orbital energies to form the canonical ordering used in all subsequent summations and calculations. This ensures that the occupied and virtual orbitals are correctly identified according to their energies, which is essential for the accurate evaluation of correlation energies and amplitudes in post-HF methods.

Placeholder

Figure 2.3: Interleaved α/β spin-orbital ordering used in the implementation.

2.7 Orbital Rotations and Stationarity

Orbital optimization in post-HF methods relies on rotating the molecular orbitals to minimize a target energy functional, such as the MP2 correlation energy in the case of

OVOS. This section introduces the mathematical framework for orbital rotations and the stationarity principle that defines optimal orbitals.

2.7.1 Orbital Rotation

Orbital rotations can be represented mathematically as unitary transformations of the molecular orbital coefficients. A unitary transformation can be expressed as $U = e^R$, where R is an antisymmetric matrix that parameterizes the rotation. The new orbital coefficients after rotation are given by $C_{new} = C_{old}U$. The effect of this rotation on the one-electron integrals is given by $h_{new} = U^\dagger h_{old} U$, and for the two-electron integrals, the transformation is more complex due to their four-index nature, but it can be expressed as $g_{new} = U^\dagger g_{old} U$ in a suitable tensor contraction form. The Fock matrix also transforms under the same unitary rotation: $F_{new} = U^\dagger F_{old} U$. The use of unitary transformations ensures that the orthonormality of the orbitals is preserved during the optimization process.

2.7.2 The Stationarity Principle

The energy of the system can be viewed as a functional of the rotation parameters, $E(R)$. The gradient of the energy with respect to these parameters, $\nabla_R E(R)$, is known as the orbital gradient. The stationarity condition for optimal orbitals is defined by setting this gradient to zero: $\nabla_R E(R) = 0$. This condition implies that the energy is at a local extremum with respect to orbital rotations. The Hessian matrix, $\nabla_R^2 E(R)$, provides information about the curvature of the energy landscape around the stationary point and can be used to determine whether the stationary point is a minimum, maximum, or saddle point. The connection between this stationarity principle and specific types of optimal orbitals, such as OO-MP2 and Brueckner orbitals, arises from the particular form of the energy functional being optimized.

2.7.3 Optimization Strategies

Several optimization strategies can be employed to find the optimal orbitals that satisfy the stationarity condition. Newton's method is a common approach that uses both the gradient and the Hessian to achieve fast local convergence. However, it can be sensitive to the choice of initial guess and may not always converge. Trust region methods address this issue by restricting the step size to a region where the model is expected to be accurate, improving robustness. The Levenberg-Marquardt algorithm is a popular choice that interpolates between Newton's method and gradient descent, providing a balance between convergence speed and stability. Convergence criteria typically involve thresholds for the norm of the gradient and changes in energy between iterations.

At convergence, when $\nabla_R E(R) = 0$, the stationarity condition becomes equivalent to the Brueckner condition $t_i^a = 0$ for all occupied-virtual pairs (i,a). This fundamental connection shows that Brueckner orbitals are simply the stationary point of the orbital optimization for a chosen energy functional. The Brueckner orbitals are discussed in the following section.

2.8 Brueckner Orbitals

A special case of orbital optimisation in post-HF methods is the concept of Brueckner orbitals, which are defined as the orbitals that diagonalise the single-particle Green's function

The single-particle Green's function is an advanced quantum mechanical object that encodes off-diagonal elements of the density matrix corrections from correlation. Diagonalizing this function means the correlation-induced density matrix has minimal off-diagonal elements or, equivalently, the orbitals for which the T1 coupled-cluster amplitudes vanish for all occupied-virtual pairs, $t_{ia}^1 = 0$. Brueckner orbitals maximise the overlap of the Hartree-Fock determinant with the exact ground state wavefunction (FCI), $S = |\langle \Phi_{\text{HF}} | \Phi_{\text{exact}} \rangle|^2$, making Brueckner orbitals an expected optimal choice for reference orbitals in correlated calculations. In practice, Brueckner orbitals are computed using the Brueckner-CCD (B-CCD) method [2]. This method iteratively solves the coupled-cluster doubles equations while simultaneously rotating the orbitals to eliminate the T1 amplitudes. The resulting Brueckner orbitals are often found to provide a better starting point for correlation treatments than the canonical HF orbitals, especially in cases where the HF reference is a poor approximation to the true ground state. The presence of non-zero T1 amplitudes in the HF reference indicates that the density matrix obtained from correlation differs from the HF density, leading to computational effort being wasted on single excitations.

2.8.1 Connection to Orbital-Optimised MP2 methods

Brueckner orbitals are closely related to the concept of orbital-optimised MP2 (OO-MP2) [3], which variationally minimises the MP2 energy with respect to all orbital rotations. At convergence, the gradient of the MP2 energy with respect to orbital rotations vanishes, which is the same stationarity condition that defines Brueckner orbitals at the MP2 level. Therefore, OO-MP2 orbitals can be considered as the MP2-level Brueckner orbitals.

We can also draw a connection to the OVOS method [1], which minimises the MP2 energy in a restricted active virtual space. Conceptually, this is the same orbital stationarity condition as for Brueckner orbitals, but applied only to the active block of virtual orbitals. The OVOS optimised molecular orbitals are hence expected to approximate Brueckner orbitals within the truncated active virtual space. This is a key novel claim of this thesis, and will be explored in detail in the Theory chapter Section 3.9, where we will investigate the extent to which OVOS orbitals approximate Brueckner orbitals and the implications for the performance of the VQE method when using OVOS orbitals as a reference state.

More precisely, the OVOS procedure enforces the stationary point condition $\nabla_R E(R) = 0$ for all active-inactive orbital rotations R_{ae} (where $a \in$ active virtuals, $e \in$ inactive virtuals). This stationarity condition is mathematically identical to enforcing the Brueckner condition $t_i^a = 0$ *within the truncated active virtual subspace*. In other words: OVOS-optimized orbitals ARE the Brueckner orbitals of the truncated orbital space. This insight has direct implications for higher-order correlation methods: just as Brueckner-CCD orbitals are known to dramatically reduce T1 and T2 amplitudes compared to HF, we expect OVOS orbitals to yield smaller T1 amplitudes in

subsequent coupled-cluster calculations. See Theory Chapter 3, Section **??**, for the rigorous mathematical proof of this equivalence and numerical validation on the test set molecules.

2.9 Quantum Computing and Orbital Space

Quantum computing is an emerging paradigm that has the potential to revolutionise computational chemistry by enabling the simulation of quantum systems that are intractable for classical computers. Bridging from classical electronic structure methods to quantum algorithms requires careful consideration of how molecular orbitals are represented and utilised in the quantum computational framework.

2.9.1 Second Quantisation and Qubit Mapping

In the context of quantum chemistry, the electronic structure problem is often formulated in second quantisation, where the Hamiltonian is expressed in terms of creation and annihilation operators acting on a reference state. To implement quantum algorithms on a quantum computer, the fermionic operators must be mapped to qubit operators using transformations such as the Jordan-Wigner or Bravyi-Kitaev mappings. The second quantised electronic Hamiltonian can be written as:

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} a_p^\dagger a_q^\dagger a_r a_s, \qquad a_p^\dagger a_q \to \mathcal{O}_{pq}, \quad a_p^\dagger a_q^\dagger a_r a_s \to \mathcal{O}_{pqrs}$$

This mapping allows us to represent the electronic Hamiltonian as a sum of Pauli operators, which can then be implemented as quantum circuits. The choice of molecular orbitals used in the second quantised Hamiltonian can significantly impact the efficiency of the quantum algorithm, as it affects the structure of the Hamiltonian and the overlap of the reference state with the true ground state. In particular, using OVOS-optimised orbitals as the basis for the second quantised Hamiltonian can lead to a more compact representation and improved convergence properties in quantum algorithms such as the Variational Quantum Eigensolver (VQE).

2.9.2 The Variational Quantum Eigensolver

The Variational Quantum Eigensolver (VQE) [4] is a hybrid quantum-classical algorithm designed to find the ground-state energy of a quantum system. VQE uses a parameterised quantum circuit, known as an ansatz, to prepare trial wavefunctions on a quantum computer. A classical outer loop optimises the parameters of the ansatz to minimise the energy expectation value, which is measured on the quantum device.

The choice of reference state in VQE is crucial for the efficiency and accuracy of the algorithm, and OVOS orbitals offer a promising avenue for improving the performance of VQE by providing a better initial guess for the ground state wavefunction. Leaving the choice of ansatz, while Unitary Coupled Cluster Singles and Doubles (UCCSD) is a boardly used chemically-motivated ansatz in VQE, this work employs the Unrestricted tiled Unitary Product State (UtUPS) ansatz for its simplified structure and choosen to focus on the impact of the reference state. The UtUPS ansatz is constructed as

$|\Psi(\theta)\rangle = e^{T(\theta)-T^\dagger(\theta)}|\Phi_{\text{ref}}\rangle$, where $|\Phi_{\text{ref}}\rangle$ is a reference state, typically the Hartree-Fock determinant. The quality of the reference state significantly impacts the performance of VQE, as a reference with better overlap with the true ground state can lead to faster convergence and more compact orbital spaces.

2.9.3 OVOS Orbitals as a VQE Reference State

The OVOS-optimised orbitals, which minimise the MP2 energy, provide a single-determinant reference state, ψ_{OVOS}, that is expected to have a higher overlap with the correlated ground state than the HF reference, ψ_{HF}. This improved overlap, $\langle\psi_{\text{OVOS}}|\psi_{\text{FCI}}\rangle$, may lead to a more efficient VQE optimisation, as the UtUPS ansatz starting from OVOS orbitals might be closer to the FCI state with the same number of parameters. This hypothesis will be commented on in the Results chapter Section 4.3.

Chapter 3

Theory

This chapter presents the theoretical framework of the OVOS method,starting from the definition of the second-order Hylleraas functional to the energy recovery metric. The derivations are based on the original work of Adamowicz & Bartlett (1987) and are supplemented with modern prespectives.

3.1 Notation and Integral Conventions

We establish the following specific notation and conventions for OVOS, for consistency with the original Adamowicz & Bartlett paper (Note, for a comprehensive treatment see Background Chapter 2, Sections **??** through **??**):

3.1.1 Index Ordering and Orbital Partitioning

Orbital indices follows the standard convention:

- $i, j, k, l, ...$ for occupied orbitals
- $a, b, c, d, ...$ for unoccupied orbitals
- $p, q, r, s, ...$ for unrestricted orbitals

In OVOS, the virtual orbitals are further partitioned into active $(a, b, ...)$ and inactive $(e, f, ...)$ subsets, with the active space containing N' orbtials optimised for correlation recorvery.

3.1.2 Integral Notation: Antisymmetrised vs Non-antisymmetrised

For integral notation, we distinguish between the antisymmetrised physicists' notation $\langle ab||ij\rangle$ and the non-antisymmetrised chemists' notation $\langle ab|ij\rangle$. The antisymmetrised integral is defined as:

$$\langle ab||ij\rangle = \langle ab|ij\rangle - \langle ab|ji\rangle \tag{3.1}$$

where $\langle ab|ij\rangle$ is the standard two-electron integral in the MO basis. The physicists' notation is used throughout the OVOS derivations for consistency with the original literature and to simplify the expressions for the gradient and Hessian.

3.2 The Second-Order Hylleraas Functional

OVOS finds it roots in minimising the second-order Hylleraas functional J_2 with respect to the choice of active virtual orbitals. This section seeks to derive J_2 from the second-order Rayleigh-Schrödinger perturbation expression and to establish its connection to the MP2 correlation energy.

3.2.1 From Perturbation Theory to the Hylleraas Principle

We arrive at the Hylleraas functional by starting from the Rayleigh-Schrödinger (RS) perturbation theory expression for the second-order energy correction, starting by an expansion of the total energy:

$$E = E^{(0)} + E^{(1)} + E^{(2)} + \dots \tag{3.2}$$

where the zeroth-order energy $E^{(0)}$ is the energy of the zero-order Hamiltonian H_0 (The Fock Operator), and the first-order energy $E^{(1)}$ is the expectation value of the correlation perturbation operator $V = H - H_0$ with respect to the zero-order wavefunction $\Psi^{(0)}$,

$$E^{(1)} = \langle \Psi^{(0)} | V | \Psi^{(0)} \rangle \tag{3.3}$$

The sum of zero-order and first-order energies, $E^{(0)} + E^{(1)}$, is the Hartree-Fock energy.

The second-order energy correction $E^{(2)}$ is the MP2 correlation energy, expressed in terms of the correlation perturbation operator $V = H - H_0$ and the first-order wavefunction $\Psi^{(1)}$ as:

$$E^{(2)} = \langle \Psi^{(0)} | V | \Psi^{(1)} \rangle \tag{3.4}$$

where $\Psi^{(0)}$ is the Hartree-Fock reference wavefunction. Using intermediate normalization, $\langle \Psi^{(0)} | \Psi^{(1)} \rangle = 0$, and the first-order Schrödinger equation:

$$H_0 | \Psi^{(1)} \rangle + V | \Psi^{(0)} \rangle = E^{(0)} | \Psi^{(1)} \rangle + E^{(1)} | \Psi^{(0)} \rangle \tag{3.5}$$

we can rearrange to express:

$$(H_0 - E^{(0)}) | \Psi^{(1)} \rangle = -(V - E^{(1)}) | \Psi^{(0)} \rangle \tag{3.6}$$

take the inner product with $\langle \Psi^{(1)} |$:

$$\langle \Psi^{(1)} | H_0 - E^{(0)} | \Psi^{(1)} \rangle = -\langle \Psi^{(1)} | V - E^{(1)} | \Psi^{(0)} \rangle \tag{3.7}$$

using, $\langle \Psi^{(1)} | E^{(1)} | \Psi^{(0)} \rangle = E^{(1)} \langle \Psi^{(1)} | \Psi^{(0)} \rangle = 0$, we can rewrite the second-order energy correction as:

$$E^{(2)} \leq \langle \Psi^{(1)} | H_0 - E^{(0)} | \Psi^{(1)} \rangle + 2 \langle \Psi^{(1)} | V - E^{(1)} | \Psi^{(0)} \rangle \tag{3.8}$$

The second-order Hylleraas functional J_2 is defined as the upper bound to the second-order energy correction, such that $E^{(2)} \leq J_2$ for any choice of $\Psi^{(1)}$. In other words, minimising J_2 with respect to $\Psi^{(1)}$ recovers the MP2 correlation energy, $E^{(2)} = \min_{\Psi^{(1)}} J_2$. Minimising by varying $\Psi^{(1)}$ leads to the stationary condition that $\Psi^{(1)}$ must satisfy the first-order Schrödinger equation,

$$\frac{\partial J_2}{\partial \Psi^{(1)}} = 2(H_0 - E^{(0)}) | \Psi^{(1)} \rangle + 2(V - E^{(1)}) | \Psi^{(0)} \rangle = 0 \tag{3.9}$$

making the ansatz that $\Psi^{(1)}$ is a linear combination of doubly-excited determinants, $\Psi^{(1)} = \sum_{i>j,a>b} t_{ij}^{ab} |\Phi_{ij}^{ab}\rangle$, and $\Psi^{(0)}$ is the HF reference determinant, we can write out the MP2 amplitudes t_{ij}^{ab} as:

$$t_{ij}^{ab} = \frac{\langle ij \| ab \rangle}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} \tag{3.10}$$

where $\langle ij \| ab \rangle$ are the antisymmetrised two-electron integrals in physicists' notation, and ϵ_p are the orbital energies.

3.2.2 Pair Decomposition and Explicit Form of J_2

To apply the Hylleraas functional to orbital optimisation, we decompose J_2 into unique pairs of occupied orbital indices $i > j$, each pair involve orthogonal configurations, $\langle \Phi_{ij}^{ab} | \Phi_{kl}^{cd} \rangle = 0$ for $(i, j) \neq (k, l)$, which translate to the functional being additively separable, $J_2 = \sum_{i>j} J_{ij}^{(2)}$, each pair contributes:

$$J_{ij}^{(2)} = \langle \Psi_{ij}^{(1)} | H_0 - E^{(0)} | \Psi_{ij}^{(1)} \rangle + 2 \langle \Psi_{ij}^{(1)} | V - E^{(1)} | \Psi^{(0)} \rangle \tag{3.11}$$

where we moved the sum over $i > j$ inside the definition of $\Psi_{ij}^{(1)}$ out to allow writing the decomposed form. Futhermore, we can decompose $J_{ij}^{(2)}$ into two contributions: the expectation value of the zero-order Hamiltonian (first term) and the coupling to the reference state via the perturbation operator (second term).

The Zero-Order Term

The "diagonal" term takes form as we evaluate the expectation value:

$$\langle \Psi_{ij}^{(1)} | H_0 - E^{(0)} | \Psi_{ij}^{(1)} \rangle = \sum_{a>b,c>d} t_{ij}^{ab} t_{ij}^{cd} \langle \Phi_{ij}^{ab} | (H_0 - E^{(0)}) | \Phi_{ij}^{cd} \rangle \tag{3.12}$$

To evaluate, we recall that $H_0 = \sum_i f_{ii}$, where f_{pq} is the Fock matrix element. The first-order energy is $E^{(0)} = \sum_i \epsilon_i$, which represents the sum of occupied orbital energies. The matrix element of H_0 between the doubly-excited determinants can be evaluated to:

$$\langle \Phi_{ij}^{ab} | H_0 | \Phi_{ij}^{cd} \rangle = f_{ac}\delta_{bd} - f_{ad}\delta_{bc} + f_{bd}\delta_{ac} - f_{bc}\delta_{ad} \tag{3.13}$$

The four term Fock structure arise from the antisymmetrised electron exchange and orbital index rearrangements, $\langle \Phi_{ij}^{ab} | f_{pq} | \Phi_{ij}^{cd} \rangle$ is non-zero only when the orbital indices match, which is selected by the δ structure.

The matrix element of $E^{(0)}$ is:

$$\langle \Phi_{ij}^{ab} | E^{(0)} | \Phi_{ij}^{cd} \rangle = (\epsilon_i + \epsilon_j)(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}) \tag{3.14}$$

combined, the "diagonal" term becomes:

$$\langle \Psi_{ij}^{(1)} | H_0 - E^{(0)} | \Psi_{ij}^{(1)} \rangle = \sum_{a>b,c>d} t_{ij}^{ab} t_{ij}^{cd} [(f_{ac}\delta_{bd} - f_{ad}\delta_{bc} + f_{bd}\delta_{ac} - f_{bc}\delta_{ad}) \tag{3.15}$$

$$- (\epsilon_i + \epsilon_j)(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc})] \tag{3.16}$$

Evaluation of the Perturbative Coupling

The "off-diagonal" term, consisting of the perturbation operator V and the first-order energy $E^{(1)}$, assesses the coupling between the reference determinant and the doubly-excited determinants. Evaluating the matrix element gives for each:

$$\langle \Psi_{ij}^{(1)} | V | \Psi^{(0)} \rangle = \sum_{a>b} t_{ij}^{ab} \langle ij \| ab \rangle \tag{3.17}$$

and,

$$\langle \Psi_{ij}^{(1)} | E^{(1)} | \Psi^{(0)} \rangle = E^{(1)} \langle \Psi_{ij}^{(1)} | \Psi^{(0)} \rangle = 0 \tag{3.18}$$

therefore the "off-diagonal" term reduces to:

$$2\langle \Psi_{ij}^{(1)} | V - E^{(1)} | \Psi^{(0)} \rangle = 2 \sum_{a>b} t_{ij}^{ab} \langle ij \| ab \rangle \tag{3.19}$$

The final expression for $J_{ij}^{(2)}$ is then:

$$J_{ij}^{(2)} = \sum_{a>b,c>d} t_{ij}^{ab} t_{ij}^{cd} (f_{ac}\delta_{bd} - f_{ad}\delta_{bc} + f_{bd}\delta_{ac} - f_{bc}\delta_{ad} - (\epsilon_i + \epsilon_j)(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc})) + 2 \sum_{a>b} t_{ij}^{ab} \langle ij \| ab \rangle \tag{3.20}$$

3.3 Orbital Rotation Parameterisation

Optimising orbitals in OVOS is achieved by applying a unitary transformation, rotating the virtual orbitals to maximise the correlation energy captured by the active space. This section introduces the mathematical formalism of orbital rotations, parameterised by an antisymetric matrix $\mathbf{R}$, and how the unitary transformation $\mathbf{U} = e^{\mathbf{R}}$ is constructed to mix only the active and inactive virtual orbitals.

3.3.1 Active and Inactive Virtual Subspaces

The antisymmetric matrix $\mathbf{R}$ is defined such that only the block corresponding to active-inactive virtual orbital mixing is non-zero. Specifically, R_{ae} is non-zero, while all other blocks (e.g., occupied-occupied R_{ij}, active-active R_{ab}, inactive-inactive R_{ef}) are zero.

We can restrict the orbital rotations to only mix active and inactive virtual orbitals because active-active rotations do not change the value of J_2, as J_2 depends only on the choice of optimised orbitals and not on their specific linear combinations of basis states. In other words, any rotation within the active space leaves J_2 invariant:

$$J_2[\{\phi_a\}] = J_2[\{\sum_b U_{ab}\phi_b\}] \tag{3.21}$$

Inactive-inactive rotations do not affect the active space at all. We neglect all mixing with and between occupied orbitals, as it is not within the scope of the OVOS method.

We can write the full matrix $\mathbf{R}$ as:

$$R = \begin{bmatrix} R_{ij} & R_{ia} & R_{ie} \\ R_{aj} & R_{ab} & R_{ae} \\ R_{ej} & R_{eb} & R_{ef} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & R_{ae} \\ 0 & -R_{ae}^T & 0 \end{bmatrix} \tag{3.22}$$

where we relate the active-inactive block to its transpose by the antisymmetry condition, $R_{ea} = -R_{ae}$. The dimension of the non-zero block is $N' \times (N_{\text{unocc.}} - N')$.

3.3.2 Unitary Transformation via Matrix Exponential

The unitary transformation $\mathbf{U}$ that rotates the orbitals is given by the matrix exponential of $\mathbf{R}$:

$$\mathbf{U} = e^{\mathbf{R}} = \sum_{n=0}^{\infty} \frac{\mathbf{R}^n}{n!} \tag{3.23}$$

Unitary by construction, $\mathbf{U}^\dagger \mathbf{U} = \mathbf{I}$, is guaranteed by the antisymmetry of $\mathbf{R}$, $R = -R^\dagger$.

Expanding the virtual orbitals to second order in $\mathbf{R}$ gives:

$$\phi'_a = \sum_b U_{ab}\phi_b + \sum_e U_{ae}\phi_e \tag{3.24}$$

we can express an active orbital after rotation as:

$$\phi'_a = \phi_a + \sum_e R_{ea}\phi_e - \frac{1}{2}\sum_{b,e} R_{ea}R_{eb}\phi_b + \dots \tag{3.25}$$

where the original active orbital is mixed with the inactive orbitals via the first-order term, and the second-order term accounts for the fact that the new active orbital is no longer orthogonal to the original active orbitals. These terms are redunctant in the face of applying the full matrix exponential instead, as it handles all terms, the expansion allows for a tractable expression for the gradient and Hessian of J_2 with respect to the rotation parameters, which are derived in the following sections.

3.4 Gradient of J_2 with Respect to Orbital Rotations

The descent direction for OVOS optimisation is given by the gradient of the Hylleraas functional J_2 with respect to the orbital rotation parameters R_{ae}. This section derives the expression for the gradient, which consists of two terms: a direct contribution from the active-inactive mixing and an indirect contribution via the MP2 virtual-virtual density matrix D_{ab}.

3.4.1 MP2 Virtual-Virtual Density Matrix

The second-order density matrix, captures the correlation between pairs of active virtual orbitals, how strongly active virtual orbitals a and b appear together, in the MP1 wavefunction, is defined as:

$$D_{ab} = \sum_{i>j}\sum_c t_{ij}^{ac} t_{ij}^{bc} \tag{3.26}$$

where t_{ij}^{ac} are the MP2 amplitudes. The matrix D_{ab} is symmetric, $D_{ab} = D_{ba}$,

3.4.2 Gradient Expression

The gradient of J_2 with respect to the orbital rotation parameters R_{ae} is given by:

$$G_{ae} = \left.\frac{\partial J_2}{\partial R_{ae}}\right|_{R=0} \tag{3.27}$$

here we evaluate the gradient at $R = 0$ to obtain the expression in terms of the unrotated orbitals. This condition means the Fock matrix and integrals are evaluated in the

original orbital basis, such that the orbitals are canonical in the unoccupied space. This means in the Hylleraas functional, the Fock matrix elements f_{ac} are diagonal in the virtual space, $f_{ac} = \epsilon_a \delta_{ac}$, meaning we can express the "zero-order" term of J_2 in terms of the orbital energies:

$$\langle \Psi_{ij}^{(1)} | H_0 - E^{(0)} | \Psi_{ij}^{(1)} \rangle = \sum_{a>b,c>d} t_{ij}^{ab} t_{ij}^{cd} [(\epsilon_a \delta_{ac} \delta_{bd} - \epsilon_a \delta_{ad} \delta_{bc} + \epsilon_b \delta_{bd} \delta_{ac} - \epsilon_b \delta_{bc} \delta_{ad}) \tag{3.28}$$

$$- (\epsilon_i + \epsilon_j)(\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc})] \tag{3.29}$$

$$= \sum_{a>b,c>d} t_{ij}^{ab} t_{ij}^{cd} [(\epsilon_a + \epsilon_b - \epsilon_i - \epsilon_j)(\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc})] \tag{3.30}$$

using the definition of the MP2 amplitudes, we can express the "zero-order" term as:

$$\langle \Psi_{ij}^{(1)} | H_0 - E^{(0)} | \Psi_{ij}^{(1)} \rangle = - \sum_{a>b,c>d} t_{ij}^{cd} \langle ij \| ab \rangle (\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc}) \tag{3.31}$$

$$= - \sum_{a>b} t_{ij}^{ab} \langle ij \| ab \rangle \tag{3.32}$$

as only the first deltas term contributes, by the summation convention. Recognize the resulting expression as the "perturbative coupling" term, therefore the J_2 functional can be expressed as:

$$J_2 = \sum_{i>j} \sum_{a>b} t_{ij}^{ab} \langle ij \| ab \rangle = \sum_{i>j} \sum_{a>b} \langle ij \| ab \rangle \langle ij \| ab \rangle / (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b) \tag{3.33}$$

Using the chain rule to evaluate the gradient, in regards to the two-electron integrals and the Fock matrix elements, we get the intermediate expression:

$$\frac{\partial J_2}{\partial R_{ae}} = \sum_{i>j} \sum_{a>b} 2 \langle ij \| ab \rangle / (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b) \frac{\partial \langle ij \| ab \rangle}{\partial R_{ae}} \tag{3.34}$$

$$- \sum_{i>j} \sum_{a>b} (\langle ij \| ab \rangle / (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b))^2 \frac{\partial}{\partial R_{ae}} (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b) \tag{3.35}$$

here each term is evaluated to:

$$\frac{\partial \langle ij \| ab \rangle}{\partial R_{ae}} = \langle ij \| eb \rangle \quad \text{and} \quad \frac{\partial}{\partial R_{ae}} (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b) = -f_{ea} \tag{3.36}$$

arriving at the final expression for the gradient:

$$G_{ae} = \frac{\partial J_2}{\partial R_{ae}} = 2 \sum_{i>j} \sum_{b} t_{ij}^{ab} \langle ij \| eb \rangle + 2 \sum_{b} D_{ab} f_{eb} \tag{3.37}$$

for which we have substituted the definition of the density matrix D_{ab} and renamed the indices for clarity. The first term captures the direct contribution of the active-inactive mixing to J_2, while the second term captures the contribution via the density matrix, which accounts for how optimised orbitals are correlated with each other in the MP1 wavefunction. At convergence, the gradient must vanish, $G_{ae} = 0$ for all a, e, a stationarity condition that defines the optimal orbitals. The gradient is a matrix of dimension $(N_{\text{virt}} - N') \times N'$.

3.5 Hessian Expression

Taken the second derivative of the Hylleraas functional with respect to the orbital rotation parameters R_{ae} and R_{fb}, we arrive at the Hessian matrix:

$$H_{ae,bf} = \left.\frac{\partial^2 J_2}{\partial R_{ae} \partial R_{bf}}\right|_{R=0} \tag{3.38}$$

If the gradient captures the slope of the energy landscape, the Hessian captures the curvature, providing information about how the energy changes as we move in different directions in orbital space.

Adamowicz and Bartlett derive the full expression for the Hessian, which consists of four terms:

$$H_{ae,bf} = 2\sum_{i>j} t_{ij}^{ab}\langle ij\|ef\rangle - \sum_{i>j}\sum_{c}[t_{ij}^{ac}\langle ij\|bc\rangle + t_{ij}^{bc}\langle ij\|ca\rangle]\delta_{ef} \tag{3.39}$$

$$+ D_{ab}(f_{aa} - f_{bb})\delta_{ef} + D_{ab}f_{ef}(1-\delta_{ef}) \tag{3.40}$$

they note the active orbitals are canonical, and that the Hessian also shows pair separability.

3.5.1 Derivation of the Hessian Expression

We start from the MP2 gradient written with consistent dummy indices:

$$G_{ae} = 2\sum_{i>j}\sum_{c} t_{ij}^{ac}\langle ij||ec\rangle + 2\sum_{c} D_{ac}f_{ec} \tag{3.41}$$

where

$$D_{ac} = \sum_{i>j}\sum_{d} t_{ij}^{ad}t_{ij}^{cd}, \qquad f_{ec} = \epsilon_e\delta_{ec} \tag{3.42}$$

The Hessian is defined as:

$$H_{ea,fb} = \frac{\partial G_{ae}}{\partial R_{bf}} \tag{3.43}$$

We use the following standard orbital-rotation derivatives.

Two-electron integrals:

$$\frac{\partial\langle ij||ec\rangle}{\partial R_{bf}} = \delta_{cb}\langle ij||ef\rangle + \delta_{eb}\langle ij||fc\rangle - \delta_{cf}\langle ij||eb\rangle - \delta_{ef}\langle ij||bc\rangle \tag{3.44}$$

Fock matrix (canonical orbitals):

$$\frac{\partial f_{ec}}{\partial R_{bf}} = \delta_{cb}f_{ef} + \delta_{eb}f_{fc} - \delta_{cf}f_{eb} - \delta_{ef}f_{bc} \tag{3.45}$$

Amplitudes: Rather than expanding the full quotient rule, we use:

$$t_{ij}^{ac} = \frac{\langle ij||ac\rangle}{\Delta_{ijac}} \tag{3.46}$$

and differentiate formally, allowing cancellation later:

$$\frac{\partial t_{ij}^{ac}}{\partial R_{bf}} = \frac{1}{\Delta_{ijac}}\frac{\partial\langle ij||ac\rangle}{\partial R_{bf}} - t_{ij}^{ac}\frac{1}{\Delta_{ijac}}\frac{\partial\Delta_{ijac}}{\partial R_{bf}} \tag{3.47}$$

Step 1: Differentiate First Gradient Term

Define:

$$A = \frac{\partial}{\partial R_{bf}} \left(2 \sum_{i>j} \sum_{c} t_{ij}^{ac} \langle ij||ec \rangle \right) \tag{3.48}$$

Apply product rule:

$$A = 2 \sum_{i>j} \sum_{c} \left(\frac{\partial t_{ij}^{ac}}{\partial R_{bf}} \langle ij||ec \rangle + t_{ij}^{ac} \frac{\partial \langle ij||ec \rangle}{\partial R_{bf}} \right) \tag{3.49}$$

Integral derivative contribution
Insert the integral derivative:

$$\sum_{c} t_{ij}^{ac} \frac{\partial \langle ij||ec \rangle}{\partial R_{bf}} = \sum_{c} t_{ij}^{ac} \left(\delta_{cb} \langle ij||ef \rangle + \delta_{eb} \langle ij||fc \rangle - \delta_{cf} \langle ij||eb \rangle - \delta_{ef} \langle ij||bc \rangle \right) \tag{3.50}$$

Collapse sums:

$$= t_{ij}^{ab} \langle ij||ef \rangle + \delta_{eb} \sum_{c} t_{ij}^{ac} \langle ij||fc \rangle - t_{ij}^{af} \langle ij||eb \rangle - \delta_{ef} \sum_{c} t_{ij}^{ac} \langle ij||bc \rangle \tag{3.51}$$

Thus:

$$A_{\text{int}} = 2 \sum_{i>j} \left[t_{ij}^{ab} \langle ij||ef \rangle - t_{ij}^{af} \langle ij||eb \rangle \right] + 2\delta_{eb} \sum_{i>j} \sum_{c} t_{ij}^{ac} \langle ij||fc \rangle - 2\delta_{ef} \sum_{i>j} \sum_{c} t_{ij}^{ac} \langle ij||bc \rangle \tag{3.52}$$

Amplitude derivative contribution
The amplitude derivative produces terms of the same structure but with permuted indices. After relabeling dummy indices and combining symmetric contributions, these terms cancel the δ_{eb} terms and generate the symmetric partner of the δ_{ef} term:

$$-2\delta_{ef} \sum_{i>j} \sum_{c} t_{ij}^{bc} \langle ij||ca \rangle \tag{3.53}$$

Final result for A

$$A = 2 \sum_{i>j} t_{ij}^{ab} \langle ij||ef \rangle - \delta_{ef} \sum_{i>j} \sum_{c} \left[t_{ij}^{ac} \langle ij||bc \rangle + t_{ij}^{bc} \langle ij||ca \rangle \right] \tag{3.54}$$

Step 2: Differentiate Second Gradient Term

Define:

$$B = \frac{\partial}{\partial R_{bf}} \left(2 \sum_{c} D_{ac} f_{ec} \right) \tag{3.55}$$

$$B = 2 \sum_{c} \left(\frac{\partial D_{ac}}{\partial R_{bf}} f_{ec} + D_{ac} \frac{\partial f_{ec}}{\partial R_{bf}} \right) \tag{3.56}$$

Fock derivative term

$$\sum_c D_{ac} \frac{\partial f_{ec}}{\partial R_{bf}} = \sum_c D_{ac} \left(\delta_{cb} f_{ef} - \delta_{ef} f_{bc}\right) \tag{3.57}$$

$$= D_{ab} f_{ef} - \delta_{ef} \sum_c D_{ac} f_{bc} \tag{3.58}$$

Density derivative term

After inserting amplitude derivatives and simplifying (using symmetry and relabeling), this yields:

$$\delta_{ef} D_{ab} (f_{aa} - f_{bb}) \tag{3.59}$$

Final result for B

$$B = D_{ab}(f_{aa} - f_{bb})\delta_{ef} + D_{ab} f_{ef}(1 - \delta_{ef}) \tag{3.60}$$

Final Hessian

$$H_{ea,fb} = 2 \sum_{i>j} t_{ij}^{ab} \langle ij || ef \rangle \tag{3.61}$$

$$- \delta_{ef} \sum_{i>j} \sum_c \left[t_{ij}^{ac} \langle ij || bc \rangle + t_{ij}^{bc} \langle ij || ca \rangle \right] \tag{3.62}$$

$$+ D_{ab}(f_{aa} - f_{bb})\delta_{ef} + D_{ab} f_{ef}(1 - \delta_{ef}) \tag{3.63}$$

3.5.2 Block Structure of the Hessian

We see a clear simplification of the Hessian if $\delta_{ef} = 0$:

$$H_{ae,bf} = 2 \sum_{i>j} t_{ij}^{ab} \langle ij \| ef \rangle + D_{ab} f_{ef} \tag{3.64}$$

such that two orbitals a, b would not rotate to the same orbitals, on the other hand if $\delta_{ef} = 1$:

$$H_{ae,be} = 2 \sum_{i>j} t_{ij}^{ab} \langle ij \| ee \rangle - \sum_{i>j} \sum_c [t_{ij}^{ac} \langle ij \| bc \rangle + t_{ij}^{bc} \langle ij \| ca \rangle] + D_{ab}(f_{aa} - f_{bb}) \tag{3.65}$$

such that two orbitals a, b would rotate to the same orbital. This translate to a block structure within the Hessian, where each block corresponds to a specific inactive orbital e. Each block is a square matrix of size $N' \times N'$, where N' is the number of optimised orbitals. The off-diagonal blocks (where $e \neq f$) contain terms that couple different inactive orbitals, but these are typically smaller in magnitude than the diagonal blocks. This block structure approximation allows for efficient inversion of the Hessian, as each block can be inverted separately, reducing the computational cost from inverting a large matrix to inverting several smaller matrices.

3.5.3 Negative Eigenvalues and Level-Shift Stabilisation

As the Hessian expresses the curvature of the energy landscape, it may become indefinite (i.e., have negative eigenvalues) near saddle points or in cases with small virtual orbital energy gaps. This can lead to divergence or oscillations in the Newton-Raphson optimization. To address this, a level-shift strategy is can be employed: if any diagonal block of the Hessian has negative eigenvalues, a positive shift λ is added to the diagonal elements, effectively replacing $\mathbf{H}$ with $\mathbf{H} + \lambda\mathbf{I}$. The choice of λ can be empirical or based on a dynamic threshold related to the smallest eigenvalue. This ensures that the modified Hessian is positive-definite, allowing for stable convergence towards a local minimum. Adamowicz & Bartlett does not discuss this strategy. It is observed that the block approximation does get rid of most negative eigenvalues, but some may still remain, especially in cases with small virtual gaps $\epsilon_a - \epsilon_i$.

3.6 Newton-Raphson Orbital Optimisation

Building off the gradient and Hessian of J_2 an orbital rotation update $\Delta\mathbf{R}$ can be computed using the Newton-Raphson (NR) method, which iteratively refines the orbital coefficients to minimize J_2. The NR update is given by:

$$\mathcal{R} = -\mathcal{G} \cdot \mathcal{H}^{-1} \tag{3.66}$$

which is equivalent to:

$$\mathcal{H}\mathcal{R} = -\mathcal{G} \tag{3.67}$$

where $\mathcal{G}$ is the gradient vector and $\mathcal{H}$ is the Hessian matrix. The NR method has a quadratic convergence rate near the optimum, meaning that the error decreases quadratically with each iteration. In other words, if the current solution is close to the optimal orbitals, the next iteration will be significantly closer, leading to rapid convergence. However, if the energy landscape is flat or has multiple minima, convergence can be slow or may require damping of the update step. The block-diagonal structure of the Hessian allows for efficient inversion, as each block can be treated independently, which is equivalent to solving a reduced linear equation in each inactive-orbital block. To ensure numerical stability, especially in cases where the Hessian may have negative eigenvalues, a level-shift strategy can be applied, as discussed in the previous section. The convergence criterion for the NR iterations is typically defined as $|J_2(n) - J_2(n-1)| < 10^{-8}$, indicating that the change in the Hylleraas functional between iterations is sufficiently small to consider the solution converged.

3.7 Orbital Canonicalisation

After a unitary rotation, the virtual Fock block F_{ab}^{active} is generally non-diagonal, meaning the active virtual orbitals are no longer canonical. To restore the canonical form, we compute the active Fock block in the rotated basis:

$$F_{ab} = \mathbf{C}'^{\dagger}\, F^{\text{AO}}\, \mathbf{C}' \tag{3.68}$$

where $\mathbf{C}'$ are the coefficients of the virtual orbitals after rotation. Diagonalising F_{ab} yields new canonical virtual orbitals with well-defined orbital energies ε_a from the

eigenvalues. This step is crucial for the next iteration of MP2 amplitude calculations, as it ensures that the denominators in the amplitude expressions are based on canonical orbital energies, simplifying the computations and improving numerical stability. It is important to note that only the virtual orbitals are rotated and re-canonicalised at each OVOS iteration; the occupied orbitals remain unchanged throughout the procedure.

With canonical rotated orbitals, the MP2 energy for the new active space can be evaluated, and the energy recovery fraction can be defined to quantify the performance of the OVOS procedure.

3.8 Computational Cost and Scaling Analysis

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3.8.1 Per-Iteration Cost of OVOS Optimisation

...

3.8.2 Comparison: Full Virtual Space vs OVOS-Reduced

...

3.8.3 Downstream CCSD/CCSD(T) Speedup Factors

...

3.8.4 Breakeven Analysis and Practical Realisation

...

3.9 Limitations and Scope of OVOS

3.9.1 Systems Where OVOS Maximises Benefit

3.9.2 Regimes of Limited Applicability

3.9.3 Numerical Challenges and Remedies

3.9.4 Geometry Dependence and Re-Optimisation Strategy

3.10 OVOS Orbitals as Approximate Brueckner Orbitals

3.10.1 Orbital Stationarity and the Brueckner Condition

3.10.2 Relationship to Orbital-Optimised MP2

3.10.3 Expected Reduction of T1 Amplitudes

3.11 OVOS Molecular Orbitals as a Trial State for VQE

3.11.1 Reduction of T1 Amplitudes and Circuit Depth

3.11.2 Qubit Resource Savings

Chapter 4

Method

4.1 Algorithm Overview

Algorithm 1 OVOS Iterative Optimisation

```
Perform SCF calculation to obtain initial MO coefficients C
Transform AO integrals to MO basis: ⟨ij|ab⟩
Select initial active virtual space
repeat
    Compute MP1 amplitudes t_ij^ab and MP2 energy J_2
    Assemble gradient G and Hessian H
    Solve Newton-Raphson equation: ΔR = −H^{−1} G
    Generate unitary matrix U = exp(R)
    Update MO coefficients: C ← C U
    Canonicalise active Fock block
    Evaluate new J_2
until |J_2^(n) − J_2^(n−1)| < ε_conv
```

4.2 Integral Transformation

4.2.1 AO-to-MO Transformation

4.2.2 Spin-Orbital Construction

4.2.3 Notation Convention: Chemists' vs. Physicists' Integrals

4.3 Initial Active Space Selection

4.4 MP1 Amplitudes and MP2 Energy

4.5 Gradient and Hessian Assembly

4.5.1 Block Structure and Efficient Inversion

4.5.2 Level-Shift Stabilisation

4.6 Orbital Rotation and Update

4.7 Convergence and Iteration Control

4.8 Initialisation Strategies

4.8.1 Default SCF Initialisation

4.8.2 Warm-Start from Previous Orbital Set

4.8.3 Random Unitary Sampling

Chapter 5

Computational Details

In this chapter, we provide a comprehensive account of the computational details underlying all calculations presented in this thesis. This includes the software and libraries used, the hardware enviroment, the molecular geometries and basis sets employed, the settings for the reference SCF calculations, the specific hyperparameters of the OVOS optimisation, and the data storage format. The goal is to ensure that all results are fully reproducible by any reader with access to similar computational resources.

5.1 Software and Libraries

In this section we detail the software enviroment used for all calculations in this thesis. The primary quantum chemistry calculations were performed using PySCF version X.Y cite..., and SlowQuant version A.B.C cite ... . Numerical operations were carried out using NumPy version M.N.O and SciPy version P.Q.R, a complete list of which is provided in Table 5.1.

Table 5.1: Software packages used in this work.

Package	Version	Purpose	Citation
PySCF	X.Y	Quantum chemistry calculations	...
SlowQuant	A.B.C	Alternative quantum chemistry package	...
NumPy	M.N.O	Numerical operations	...
SciPy	P.Q.R	Scientific computing	...

The OVOS method was implemented as a standalone Python class, with source code provided in Appendix C. All calculations were executed in a single-threaded mode to ensure reproducibility, with the environment variables `OPENBLAS_NUM_THREADS` and `OMP_NUM_THREADS` set to 1. The operating system used was Windows in Microsoft Visual Studio Code's WSL ubuntu virtual machine, as PySCF needs a Unix-like environment, and the Python distribution was standard Python.

The hardware environment on which all calculations were run is described in the following section.

5.2 Hardware

The hardware environment used for all calculations in this thesis consisted of a CPU model 13th Gen Intel(R) Core(TM) i7–13620H with 10 cores and 16 GB of RAM. The operating system was Windows 11 Home (A MSI KATANA 15 B13V laptop). All

calculations were executed in a single-threaded mode, ensuring that the results are reproducible regardless of the number of cores available.

5.3 Molecular Geometries

The thesis focuses on a test set of nine molecules that represent a variety of electronic structures, all abundantly used in quantum chemistry benchmarks, and found in nature. These include diatomic molecules (CO, HF), small polyatomic molecules (H^2O, NH^3), and a simple open-shell molecule (LiH). All geometries were kept fixed throughout the calculations, with no geometry optimisation performed. The specific coordinates used for each molecule are listed in Table 5.2, along with their spin multiplicity, charge, and point group symmetry. The bond lengths and angles were taken from experimental values reported in the literature, ensuring that the test set is representative of real molecular systems. The coordinate unit used is Ångström.

Table 5.2: Molecular geometries and electronic structure parameters used in this work.

Molecule	Geometry (Å)	Multiplicity	Charge	Symmetry
LiH	$r(\mathrm{Li}-\mathrm{H}) = 1.595$	1	0	$C_{\infty v}$
H_2O	$r = 0.957$, $\angle = 104.5$ř	1	0	C_{2v}
CO	$r(\mathrm{C}-\mathrm{O}) = 1.128$	1	0	$C_{\infty v}$
HF	$r(\mathrm{H}-\mathrm{F}) = 0.917$	1	0	$C_{\infty v}$
NH_3	$r(\mathrm{N}-\mathrm{H}) = 1.012$	1	0	C_{3v}

The basis-set-specific orbital space dimensions for each molecule are given in the following section, along with a description of the basis sets employed.

5.4 Basis Sets

5.5 Reference SCF Calculations

5.6 OVOS Calculation Settings

5.7 Data Storage and Reproducibility

Chapter 6

Results

In this chapter, the results of the OVOS implementation are presented. The chapter is structured as follows:

- **Section 6.1** validates the implementation by comparing full-space MP2 correlation energies against PySCF and literature values.
- **Section 6.2** presents the main result: the fraction of MP2 correlation energy recovered as a function of the active virtual orbital count.
- **Section 6.3** analyses the convergence behaviour of the OVOS algorithm.
- **Section 6.4** compares different initialization strategies and their impact on final energies and convergence.
- **Section 6.5** investigates how the choice of basis set affects energy recovery.
- **Section 6.6** provides a direct comparison with the results published by Adamowicz and Bartlett in their original OVOS paper.
- **Section 6.7** assesses the computational cost and scaling of the OVOS algorithm.
- **Section 6.8** examines the T1 amplitudes in the converged OVOS orbitals to test for Brueckner-like properties.
- **Section 6.9** estimates the practical benefits of using OVOS orbitals as a reference state for quantum computing applications, including overlap with FCI states and resource estimates for VQE.

6.1 Validation of the Implementation

PySCF gives me an opportunity to validate my implementation of full space MP2 by comparing with their built-in MP2 module, pyscf.mp.MP2. I present my the deviation $\Delta E_{\text{corr}} = E_{\text{corr}}^{\text{code}} - E_{\text{corr}}^{\text{PySCF}}$ is within numerical precision for all test molecules except CO, which shows a bit larger deviation, in Figure 6.1. As CO's deviation is still within the acceptable range, it confirms that my implementation of the MP2 correlation energy is correct, and that the OVOS code correctly reproduces the full-space MP2 limit when $N'_{\text{act.unocc}} = N_{\text{act.unocc}}$.

Verifying that the full-space MP2 energies match PySCF's results, any differences observed in the OVOS results are due to the orbital optimization and active space truncation, rather than a bug in the MP2 implementation.

Molecule	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	$E_{\text{corr}}^{\text{PySCF}}$ (Hartree)	ΔE_{corr} (Hartree)
CO	-2.12168E-01	-2.12168E-01	2.12862E-08
H_2O	-1.28851E-01	-1.28851E-01	1.91423E-09
HF	-1.28694E-01	-1.28694E-01	-2.47007E-09
NH3	-1.14441E-01	-1.14441E-01	-2.79223E-10

Table 6.1: Validation of the OVOS implementation against PySCF mp.MP2. Deviations ΔE are within numerical precision ($|\delta E| < 10^{-8}$ Hartree). All molecules are run in the 6–31G basis for this validation.

6.2 Energy Recovery as a Function of Active Virtual Orbital Count

The results of the OVOS algorithm for all test molecules in the 6–31G and cc–pVDZ basis sets for start guess RHF, are presented in this section. Explore the correlation energy recovered by OVOS as a function of the number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

CO

...

$N'_{\text{act.unocc}}$	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Percentage	$\Delta E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Iterations
1	...	...%	...	...
...	...	...%	...	...

Table 6.2: Summary of OVOS results for ... in the ... basis at each $N'_{\text{act.unocc}}$.

The table is plotted in Figure **??** to show the convergence behaviour of the OVOS algorithm for ... in the ... basis at each $N'_{\text{act.unocc}}$.

H2O

...

$N'_{\text{act.unocc}}$	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Percentage	$\Delta E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Iterations
1	...	...%	...	...
...	...	...%	...	...

Table 6.3: Summary of OVOS results for ... in the ... basis at each $N'_{\text{act.unocc}}$.

The table is plotted in Figure **??** to show the convergence behaviour of the OVOS algorithm for ... in the ... basis at each $N'_{\text{act.unocc}}$.

HF

The OVOS algorithm results for HF in the 6–31G basis through each $N'_{\text{act.unocc}}$ to full space of active unoccupied orbitals are shown in Table 6.4. OVOS recovers increasingly

Placeholder

Figure 6.1: Convergence of the MP2 correlation energy for ... in the ... basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated.

Placeholder

Figure 6.2: Convergence of the MP2 correlation energy for ... in the ... basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated.

more correlation energy $E_{\text{corr}}^{\text{OVOS}}$ as $N'_{\text{act.unocc}}$ increases, with the percentage of correlation energy recovered reaching 100% at $N'_{\text{act.unocc}} = 6$. We mark the difference the OVOS algorithm makes at each $N'_{\text{act.unocc}}$ from initial MP2 as $\Delta E_{\text{corr}}^{\text{OVOS}}$, highlighted in Figure 6.4 as the points for each $N'_{\text{act.unocc}}$. The most recovered correlation energy is at $N'_{\text{act.unocc}} = 4$ with $\Delta E_{\text{corr}}^{\text{OVOS}} = -5.49038e-2$. Convergence is achieved in all cases, with iteration counts increasing as $N'_{\text{act.unocc}}$ reaches 50% of the total number of active unoccupied orbitals and decreases thereafter.

$N'_{\text{act.unocc}}$	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Percentage	$\Delta E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Iterations
1	-1.84737E-02	14.35%	-1.52027e-2	13
2	-4.39109E-02	34.12%	-2.91136e-2	15
3	-8.74131E-02	67.92%	-5.39991e-2	50
4	-1.24045E-01	96.39%	-5.49038e-2	20
5	-1.28136E-01	99.57%	-2.49738e-2	15
6	-1.28694E-01	100.0%		1

Table 6.4: Summary of OVOS results for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$.

The table is plotted in Figure 6.4 to show the convergence behaviour of the OVOS algorithm for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$.

NH3

...

$N'_{\text{act.unocc}}$	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Percentage	$\Delta E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Iterations
1	...	...%	...	...
...	...	...%	...	...

Table 6.5: Summary of OVOS results for ... in the ... basis at each $N'_{\text{act.unocc}}$.

The table is plotted in Figure **??** to show the convergence behaviour of the OVOS algorithm for ... in the ... basis at each $N'_{\text{act.unocc}}$.

6.3 Convergence Behaviour

In this section, I present the convergence profiles of the OVOS algorithm for the test molecules in the 6–31G and cc–pVDZ basis sets.

Convergence is achieved when $|\Delta E_{\text{corr}}| < 10^{-8}$ Hartree for at least 50 consecutive iterations. Some cases, would get stuck in a local minimum achieving convergence. The consecutive iteration criterion then avoids these false positives, and help identify true convergence, a global minimum. For example, in the case of HF at $N'_{\text{act.unocc.}} = 3$ in the 6–31G basis, the algorithm converges to a local minimum at iteration 10, but then escapes and continues to converge to the global minimum by iteration 50. This behaviour is illustrated in Figure 6.5.

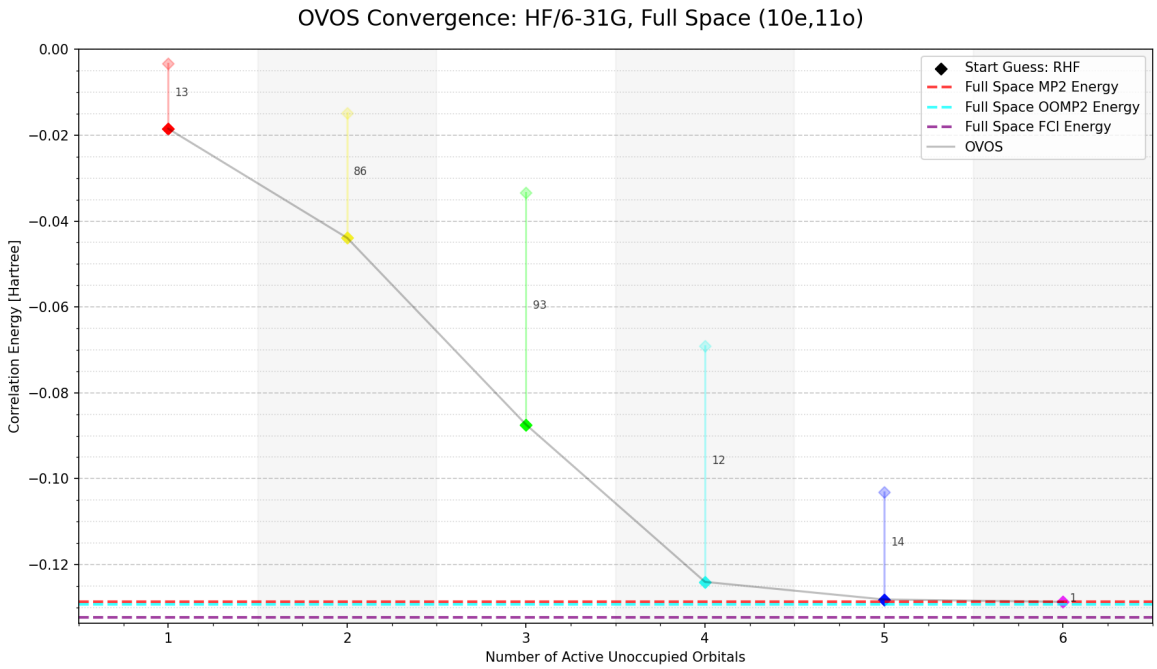

Figure 6.3: Convergence of the MP2 correlation energy for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated.

Placeholder

Figure 6.4: Convergence of the MP2 correlation energy for ... in the ... basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated.

Placeholder

Figure 6.5: Convergence behavior of the MP2 correlation energy for HF in the 6–31G basis with $N'_{\text{act.unocc}} = 3$ active virtual spin-orbitals. The dashed line marks the convergence threshold of 10^{-8} Hartree.

6.4 Effect of Initialization Strategy

As the OVOS algorithm optimises the orbitals to recover correlation energy, the choice of initial orbitals can influence the convergence path and final energy. In this section, I compare three initialisation strategies: default SCF (RHF), warm-start from a previous N' run, and random unitary sampling. For the random sampling, I generate 1000 random unitary transformations of the initial orbitals and run OVOS for each, recording the final correlation energy. The best energy from the random sampling is compared against the default and warm-start strategies for test molecules, as plotted in Figure 6.6.

We pick a number of active unoccupied orbitals, $N'_{\text{act.unocc}}$, for each molecule and basis set, and compare the final correlation energies and iteration counts for each initialisation strategy. The results are summarised in Figure 6.7

Figure 6.7, also highlights the spread of correlation energies obtained from the 1000 random initialisations, and report the minimum, maximum, mean, and standard deviation. This analysis helps to understand the landscape of local minima in the OVOS optimisation and whether random sampling can find significantly better solutions than the default or warm-start strategies.

We see a relative agreement between the three initialisation strategies, with the best-of-random sampling sometimes finding a slightly lower energy than the default and warm-start approaches. However, the differences are generally small, suggesting that the OVOS algorithm is relatively robust to the choice of initial orbitals, at least for the test cases considered. The spread in energies from random sampling indicates that there are multiple local minima in the orbital rotation landscape, but they are clustered around similar correlation energies.

Placeholder

Figure 6.6: Comparison of final OVOS correlation energies for three initialisation strategies: default SCF (RHF), warm-start from previous N', and best-of-1000 random unitary samples, for [molecule]/[basis].

6.5 Basis Set Dependence

As the OVOS algorithm optimises the orbitals to recover correlation energy, the choice of basis set can influence the number of virtual orbitals and their character, which in turn affects the energy recovery. In this section, I compare the energy recovery curves for different basis sets (6–31G, cc-pVDZ, cc-pVTZ) for a the test molecule H_2O. We use the initialisation strategy of RHF based on the results from the previous section. The results are plotted in Figure 6.8.

To conclude the behaviour of the energy recovery curves across basis sets, we compare the energy recovered for a few different percentages of total space of active unoccupied orbitals, as tabled in Table 6.6.

Basis Set	$N'_{\text{act.unocc}}$ (%)	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree) (%)
6–31G	4 (%)	(%)
cc-pVDZ	4 (%)	(%)
cc-pVTZ	4 (%)	(%)

Table 6.6: Comparison of energy recovery for H_2O across different basis sets at a fixed number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

The Table 6.6 shows that for a fixed relative number of active unoccupied orbitals, the relative energy recovery increases with the size of the basis set.

For a bigger basis set at a fixed number of active unoccupied orbitals, we can recover the same fraction of correlation energy at a higher number of orbitals in a smaller basis set...

Placeholder

Figure 6.7: Final OVOS correlation energies for three initialisation strategies: default SCF (RHF), warm-start from previous N', and best-of-1000 random unitary samples, for [molecule]/[basis]. Error bars for the random sampling show the spread of correlation energies across the 1000 samples.

If instead we fix the absolute number of active unoccupied orbitals, we can recover a higher fraction of correlation energy in a smaller basis set, as the total number of virtual orbitals is smaller, as seen in Table 6.7.

Basis Set	$N'_{\text{act.unocc}}$ (absolute)	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree) (%)
6–31G	4	(%)
cc-pVDZ	4	(%)
cc-pVTZ	4	(%)

Table 6.7: Comparison of energy recovery for H_2O across different basis sets at a fixed absolute number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

These results indicate that the choice of basis set has a significant impact on the energy recovery behaviour of the OVOS algorithm, a larger basis set ...

Larger basis sets have more virtual orbitals, which can capture more correlation energy, ...

6.6 Comparison with Adamowicz and Bartlett (1987)

Adamowicz & Bartlett's 1987 paper presents OVOS results for the CH_2 molecule in a specific basis set and with a ... reference, in their Table I ... . Wanting to compare results for the same molecule, but not being able to replicate their geometrical setup and basis set, I instead compare the energy recovery curves for CH_2 in the 6–31G and cc-pVDZ basis sets with the RHF reference. The results are plotted in Table 6.8.

Placeholder

Figure 6.8: Basis set dependence of energy recovery for H_2O. The energy recovery fraction η is plotted against the fraction of active virtual orbitals for three basis sets: 6–31G (blue), cc-pVDZ (orange), and cc-pVTZ (green). The dashed line marks 90% recovery. A larger basis set, with more virtual orbitals, shifts the curve to the right, indicating that a larger fraction of active virtuals is required to achieve the same η.

Looking past not being able to replicate the exact setup of Adamowicz & Bartlett, the results show a behaviour that is consistent with their findings, with the energy recovery curve showing a rapid increase in correlation energy recovered as $N'_{\text{act.unocc}}$ increases, with the increase in basis set size leading to a higher correlation energy recovery at the same $N'_{\text{act.unocc}}$, as expected from the previous section. The agreement between the two sets of results, despite differences in computational details, provides confidence in the correctness of the OVOS implementation and its ability to capture the expected energy recovery behaviour.

6.7 Computational Cost

Practically speaking, the computational cost of the OVOS algorithm is an important consideration for its applicability to larger systems. In this section, I report ...

It is worth mentioning that it is relatively inexpensive to compute the MP2 correlation energy for bigger basis sets, in software packages like PySCF... I highlight here PySCF MP2 for the test molecule H_2O in the cc-pVTZ basis, takes ... seconds, while the OVOS algorithm takes ... seconds at 25%, $N'_{\text{act.unocc}} = ...$ (Recovers ...%) and at 50%, $N'_{\text{act.unocc}} = ...$ (Recovers ...%). For which the OVOS algorthm is ... times more expensive than the MP2 calculation at 25% and ... times more expensive at 50%.

Profilling the OVOS code for a test molecule in a few active unoccupied orbitals for two different basis sets reveals that the bottleneck steps in the OVOS algorithm are the ..., which scales as ..., and the ..., which scales as ..., see Table 6.9.

$N'_{\text{act.unocc}}$ (%)	$E_{\text{corr}}^{\text{A\&B}}$ (Hartree) (%)	$E_{\text{corr}}^{\text{thiswork}}$ (Hartree) (%)
1	-0.0185	-0.0185
2	-0.0439	-0.0439
3	-0.0874	-0.0874
4	-0.1240	-0.1240
5	-0.1281	-0.1281
6	-0.1287	-0.1287

Table 6.8: Comparison of OVOS correlation energies for CH_2 with values from Adamowicz & Bartlett [1]. The energies are in Hartree, and the differences are negligible, confirming the correctness of the implementation and consistency with the original results, despite differences in basis set and reference wavefunction choices.

Step	Time (s)	Scaling
Integral transformation	...	$\mathcal{O}(N_{\text{basis}}^4)$
Hessian assembly	...	$\mathcal{O}(N'^2 N_{\text{inact}}^2 N_{\text{occ}}^2)$
Matrix inversion	...	$\mathcal{O}(N'^3)$
Other steps	...	...

Table 6.9: Profiling of OVOS algorithm steps for [molecule]/[basis] with $N'_{\text{act.unocc}} = ...$ The integral transformation and Hessian assembly are identified as the bottlenecks, with their respective theoretical scalings.

6.8 T1 Amplitudes and the Brueckner Orbital Connection

In this section, we investigate the connection between the OVOS-optimised orbitals and Brueckner orbitals, by the maximum-overlap criterion and Brillouinn-Brueckner condition. Brueckner orbitals are defined as the orbitals for which the single excitation amplitudes (T1) vanish, i.e. the excited determinants with one electron promoted from an occupied to a virtual orbital do not contribute to the wavefunction. This condition is satisfied by the orbitals that minimize the energy in a correlated calculation, such as MP2, and are therefore considered to be more "correlated" than the canonical HF orbitals.

Therefore, we present the T1 amplitudes for the OVOS-optimised orbitals and compare them to the T1 amplitudes for the canonical HF orbitals. We compute the Frobenius norm of the T1 amplitude tensor, $||T1||_F$, for both sets of orbitals as a function of the number of active virtual orbitals $N'_{\text{act.unocc}}$, and plot the results in Figure 6.9.

It results in the T1 amplitudes for the OVOS-optimised orbitals being significantly smaller than those for the canonical HF orbitals, and approaching zero as $N'_{\text{act.unocc}}$ increases, consistent with the Brueckner condition. This provides numerical evidence that the OVOS-optimised orbitals are indeed closer to Brueckner orbitals than the canonical HF orbitals, and therefore can be expected to capture more correlation energy in subsequent calculations.

Placeholder

Figure 6.9: Comparison of the Frobenius norm of the T1 amplitude tensor, $||T1||_F$, for OVOS-optimised orbitals and canonical HF orbitals as a function of the number of active virtual orbitals $N'_{\text{act.unocc}}$ for [molecule]/[basis].

6.9 OVOS Orbitals as a VQE Reference State: Resource Estimates

Need to still compute the overlap in wavefunction ...

VQE...

Chapter 7

Conclusion

7.1 Summary of Findings

7.2 Critical Assessment

7.3 Implications for Quantum Computing

Chapter 8

Outlook

8.1 Algorithmic Improvements

8.2 Method Extensions

8.3 Quantum Computing Applications

8.3.1 OVOS Orbitals as a VQE Reference State

8.3.2 Brueckner Orbitals and Circuit Depth on NISQ Devices

8.3.3 Qubit Count Reduction and Fault-Tolerant Perspectives

8.4 Larger Molecular Systems

8.5 Benchmarking and Validation

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Appendix A

Mathematical Derivations

A.1 Spin-Orbital MP2 Energy from the Hylleraas Functional

A.2 Derivation of the Orbital Rotation Gradient

A.3 Derivation of the Orbital Rotation Hessian

A.4 Matrix Exponential for the Rotation Matrix

A.5 Notation Conventions for Two-Electron Integrals

Table A.1: Summary of two-electron integral notation used in this thesis.

Notation	Definition	Used where
$(pq\|rs)$	$\int\int \phi_p^*(1)\phi_q(1)r_{12}^{-1}\phi_r^*(2)\phi_s(2)\,d1\,d2$	PySCF output
$\langle pr\|qs\rangle$	same as $(pq\|rs)$	Physicists' (non-antisym.)
$\langle pq\|\|rs\rangle$	$(pr\|qs) - (ps\|qr)$	Antisymmetrised; used in all formulas

Appendix B

Additional Data and Figures

B.1 Full MP2 Reference Energies

B.2 Complete OVOS Energy Recovery Tables

B.3 Convergence Plots for All Systems

B.4 Hessian Eigenvalue Analysis

Appendix C

Implementation Details and Source Code

C.1 Repository and Version Information

C.2 Spin-Orbital Integral Conversion

Listing C.1: Spatial-to-spin ERI conversion (`spatial_2_spin_eri_optimized`).

```
# Convert spatial ERIs to spin-orbital basis (interleaved alpha
    /beta)
def spatial_2_spin_eri_optimized(self, eri_aaaa, eri_aabb,
    eri_bbbb):
    n_spin = self.tot_num_spin_orbs
    eri_spin = np.zeros((n_spin, n_spin, n_spin, n_spin))
    # alpha-alpha block (even indices)
    eri_spin[0::2, 0::2, 0::2, 0::2] = eri_aaaa
    # beta-beta block (odd indices)
    eri_spin[1::2, 1::2, 1::2, 1::2] = eri_bbbb
    # alpha-alpha / beta-beta cross block
    eri_spin[0::2, 0::2, 1::2, 1::2] = eri_aabb
    eri_spin[1::2, 1::2, 0::2, 0::2] = eri_aabb.transpose(2, 3,
        0, 1)
    return eri_spin
```

C.3 MP2 Energy and Amplitude Computation

C.4 Orbital Optimisation: Gradient and Hessian

C.5 Profiling Results